

YASSER TAHA

Machine Learning R&D

Berlin, Germany

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<https://yasser-taha.github.io>

PROFILE

Ω Working in a dynamic and multidisciplinary scientific environment that fosters seamless information exchange and diverse perspectives Ω Successfully contributed to a range of interdisciplinary machine learning (ML) projects, spanning Autonomous Systems, Pharmaceutical Recommender Systems, and Early Diagnostics Ω Demonstrated a track record of propelling projects from initial research stages to tangible and impactful developments Ω Cultivated a rich perspective by residing in four different countries, embracing diverse norms and enriching life experiences.

EMPLOYMENT HISTORY

❖ **PhD Researcher, Robert Koch Institute & Charité Medical University** Oct 2024 — Present
Berlin

Leading a project to develop an AI-driven tool for identifying infected cell cultures, including those caused by viruses that produce limited or no visible CPE. This tool has the potential to increase virus isolation rates, detect unknown viruses, and speed up the isolation process. The methodology can be applied to various research areas in virus detection using imaging techniques.

Methods: ViLSTM, Vision Transformer, Conformal prediction

❖ **Machine Learning Researcher, HB-DX Diagnostics** Apr 2021 — Jul 2024
Heidelberg

Worked on Discovery of RNA biomarkers for early lung cancer detection from patients' blood biopsy resulting in 0.76 (stage I) to 0.81 (Stage IV) AUCROC.

Conceived an explainable "TransfoRNA" ML framework for mapping small non-coding RNA NGS fragments back to their parental precursor, shedding light on their functional role and involvement in disease mechanisms.

Developed a statistical data pipeline facilitating detailed comparison between fragment RPM counts (NGS) vs CPM (dPCR).

Methods: Transformers, Explainability, Contrastive Learning, Multimodality, Visualization.

❖ **Machine Learning R&D Engineer, Cognigy** Oct 2020 — Apr 2021
Düsseldorf

Developed ConveRT's state of the art language model and trained it on a response selection task based on Reddit dataset. The model was then utilized for intent detection.

Methods: NLP, Transformers, Unsupervised Learning, Generative Models

❖ **Machine Learning Researcher, STTech (TUM)** Oct 2019 — Sept 2020
Munich

Exploit the cross-selling relations between different pharmaceutical products as well as product ingredients to provide better recommendations. Devised a novel approach to re-rank edge weights in the cross-selling graph to overcome popularity bias in retail. The project is in production at Pharmatechnik.

Methods: Bayesian Statistics, GNNs, Autoencoders, Contrastive Learning, Recommender Systems

❖ **Research Assistant, Institute of Machine Learning, JKU** Aug 2018 — Sept 2019
Linz

Supervised by Professor Sepp Hochreiter Researched and implemented the state-of-the-art approach for Unsupervised Depth Estimation from monocular sequence of images. Reproduced the state-of-the-art results and added two novel losses to the framework that led to better evaluation on the Kitti dataset.

Methods: Computer Vision, Flow Nets, Unsupervised Learning, Convolutional Autoencoders

❖ **Algorithm Engineer, Autonomous Vehicles at Valeo** Jul 2016 — Aug 2017
Cairo

Implemented online external calibration for LiDAR sensor allowing vehicles to adapt more robustly to changes in sensor orientation. Designed and implemented ground detection algorithm from laser 3D point cloud. Both algorithms are currently deployed in production.

Methods: Computational Geometry, 3D Point Cloud, Kalman Filters

❖ **Teaching Assistant, German University In Cairo** May 2014 — Sept 2015
Cairo

Teaching: Sensor technology, Pneumatics and PLC.

Develop projects through research and implementation.

Aid students performing their bachelor projects.

PUBLICATIONS

❖ **Drainage: A Unifying Framework for Addressing Class Uncertainty** Dec 2025
CVPR 2026 (Highlight)

First Author, DOI: <https://doi.org/10.48550/arXiv.2512.03182>

❖ **TransfoRNA: Navigating the Uncertainties of Small RNA Annotation with an Adaptive Machine Learning Strategy** Jul 2024
Biorxiv

First Author, DOI: <https://doi.org/10.1101/2024.06.19.599329>

Website: www.transforna.com

❖ **Early Detection of Lung Cancer using small RNAs** Jun 2023
Journal of Thoracic Oncology

DOI: [10.1016/j.jtho.2023.07.005](https://doi.org/10.1016/j.jtho.2023.07.005)

❖ **Graph CNN for a Pharmacy Cross-Selling Recommender System** Nov 2020
Information

First co-author, DOI: <https://doi.org/10.3390/info11110525>

EDUCATION AND TRAINING

❖ **AI in Public Health Research Symposium** May 2025
Berlin

Poster: AI-Driven Classification of Virus Infections via Cytopathic Effect Imaging

❖ **EMBL** Apr 2024
Heidelberg

AI and Biology

❖ **ELLIS Summer School** Jun 2023
Manchester

ML for Healthcare and Biology

❖ **Conference** Apr 2023
Tübingen

Causal Learning and Reasoning

❖ **Johannes Kepler University** Sept 2017 — Jul 2018
Linz

MSc, Data Science

Thesis Title: “Trajectory Prediction in the Realm of Sparse Data”

Full Scholarship Thesis GPA: 2.0

♣ **Information Technology Institute** Oct 2015 — Jul 2016
PgDip in Embedded Systems Cairo

Full scholarship Funded by Valeo and IBM.

♣ **German University in Cairo** Sept 2008 — Feb 2014
BSc, Mechatronics Engineering Cairo

Thesis Title: Four degrees of Freedom, automated, localized, and mapped Robot.

Thesis GPA: 1.0 | Bachelor GPA: 2.0

SKILLS

Unix	Supervised/Unsupervised Learning
Python, R, C	Computer Vision
PyTorch, TensorFlow	Language Model
Kubernetes, Docker	Geometry
Artificial Intelligence	small RNA
Bayesian Statistics	Computational Biology
Probabilistic models	Teaching
Statistics	Research
Recommender Systems	Deep Learning

LANGUAGES

German	<i>C1</i>	English	<i>C2</i>
Egyptian	<i>Native speaker</i>		